

CS247 Assignment 1 Slice Viewer

How to run

```
cd Assignment_1
cmake -S . -B build
cmake --build build -j
cd build && ./assignment
```

Key	Action
1 / 2	load lobster / head dataset
A	switch axis (X to Y to Z)
W / S	next / previous slice
B	background colour
Esc	quit

Datasets: lobster $120 \times 120 \times 34$, skewed_head $184 \times 256 \times 170$.

Implementation

Geometry. One quad in NDC, built once in `setup()` (VAO + VBO + EBO, 6 floats per vertex = position + tex coord). Nothing about it changes at runtime. The fragment shader samples a single `GL_TEXTURE_3D`, so I never re-upload data when the slice or axis changes; I only change two `mat4` uniforms per frame.

Picking the slice (`buildTexMatrix`). Each vertex carries $(u, v, 0)$. A `texTransform` matrix maps it to the 3D sample location, with the fixed axis set to the current slice:

axis	plane	maps $(u, v, 0)$ to
0	YZ	(sliceX, u, v)
1	XZ	(u, sliceY, v)
2	XY	(u, v, sliceZ)

I use $(\text{slice} + 0.5) / \text{dim}$ so it samples the voxel centre, which matches `GL_LINEAR` filtering.

Aspect ratio (`buildModelMatrix`). Compare the slice aspect to the window aspect and shrink the wider side:

```
if (winAspect > sliceAspect) sx = sliceAspect / winAspect;
else                          sy = winAspect / sliceAspect;
```

`framebufferSizeCallback` updates the viewport on resize (uses framebuffer size, so retina is fine).

Shaders. Vertex shader applies model and `texTransform`; fragment shader does `texture(volume, texCoord).r` and writes it as greyscale.

Results

See `demo.mov`: loading both datasets, axis switching, slice scrubbing, and resizing without distortion.

Built and tested on Apple Silicon, OpenGL 4.1 core profile.